Fiscal Year:	FY 2025	**Task Last Updated:**	FY 05/14/2025
PI Name:	Delp, Michael Ph.D.		
Project Title:	High Altitude Spaceflight on the Bion-M2: Effects on Arterial and Venous Vessels		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	32306-0001	**Congressional District:**	2
Comments:	Previous affiliations were University of Florida (mid-2007-June 2014), West Virginia University (mid-2005 to mid-2007), and Texas A&M University (1995 to mid-2005).		
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology NNH18ZTT002N:Russian Bion-M2 Mission
Start Date:	04/20/2020	**End Date:**	04/19/2023
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Loftus, David	**Contact Phone:**	650-604-1011
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Flight Program:	Bion-M2		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Baker, John Ph.D. (Medical College of Wisconsin, Inc.) Ledermann, Thomas Ph.D. (Florida State University) Narayanan, Anand Ph.D. (Florida State University) Tarasova, Olga Ph.D. (Moscow State University, Institute for Biomedical Problems (IMBP), Russian Academy of Sciences, Moscow, Russia) Vinogradova, Olga Ph.D. (Institute for Biomedical Problems (IMBP), Moscow, Russia) Bagher, Pooneh Ph.D. (Texas A&M University System) Mao, Xiao Wen M.D. (Loma Linda University)		
Grant/Contract No.:	80NSSC20K0996		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Emerging research demonstrates that low-dose environmental and medical radiation exposure increases the risk of mortality due to ischemic coronary artery (heart attack) and cerebrovascular (stroke) disease. Despite the potential for low-dose radiation exposure during spaceflight to have substantial adverse effects on the cardiovascular system, little research has been conducted to investigate this phenomenon. This is particularly urgent as groups of nations and corporate entities are actively planning manned missions back into deep space (i.e., the Moon and Mars), where exposure to galactic comic radiation is pervasive. The present proposal seeks to address some of the fundamental questions that remain outstanding, including: 1) Does spaceflight and space radiation adversely affect cardiac, cerebral, ophthalmic and jugular vascular function, mechanics, and structure? 2) Does space radiation and weightlessness have differential effects on coronary, cerebral, and ophthalmic arteries? 3) Is the cardiovascular dysfunction induced by space radiation and weightlessness sustained over a period of recovery, potentially leading to chronic morbidity and mortality? 4) Is the putative vascular dysfunction induced by spaceflight and space radiation associated with elevations in markers of oxidative stress and inflammation? 5) Do elevations in markers of oxidative stress and inflammation occur in arteries and veins? Results from these studies will lead to a greater understanding of the cardiovascular health risks for deep space travel.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	[NOTE: Per the NASA Space Biology Program, these investigations were discontinued (Ed., 5/14/25).]
Bibliography Type:	Description: (Last Updated: 07/09/2025)